

Taehyeon Kim

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PROFILE

Third-year Computer Science specialist at the University of Toronto (GPA 3.65/4.00), focused on machine learning and system architecture. Builds full-stack systems end to end, from production backend infrastructure and billing systems to ML pipelines, models, and fraud-detection experimentation, with deliberate attention to tradeoffs rather than just working code.

EDUCATION

University of Toronto

2024 – Present

B.Sc. Computer Science Specialist · GPA: 3.65 / 4.00

TECHNICAL SKILLS

Languages: Python, Java, C, Bash, SQL

Frameworks & Tools: FastAPI, Next.js, React, PostgreSQL, Git, Linux, Auth0, Tailwind CSS, Supabase

Cloud: AWS (IAM, VPC, EC2, S3, Lambda); pursuing AWS Certified Solutions Architect Associate

Concepts: System Design, Systems Programming (processes, pipes, signals), OOP, Design Patterns, Data Structures & Algorithms, Scrum / Agile, REST APIs

AI / ML: NumPy, scikit-learn, Gemini API, LLM integration

PROJECTS & HACKATHONS

chalk-talk (AI Blackboard Tutor Research)

May 2026 – September 2026

Solo research project · github.com/Tedd618/chalk-talk

- Built a pipeline that turned 1,191 YouTube math lecture videos into a dataset matching what the teacher says with what they write, stroke by stroke, running it in parallel across 8 to 12 lab computers.
- Ran a controlled experiment (7 setups, each trained 3 times) on whether a model can use a teacher's writing and speech to predict each other: the writing helped predict the next word, the speech did not help predict the pen, and combining both in one model did more harm than good. Wrote up the findings in a report.
- Built a browser player that reads a lesson script and speaks and draws it stroke by stroke, as the base for a future tutor that plans its own lessons.

UTM Degree Explorer (Local-First Degree Planner)

May 2026

Solo project

- Built a full-stack degree planner over a fully scraped UTM Academic Calendar (2,346 courses, 164 programs), with a rule-based parser converting prerequisite text into structured ASTs at 86% coverage.
- Implemented live prerequisite validation and a React/Zustand planner UI with Supabase-backed multi-plan sync and guest mode.

Yosemite (AI Compliance & Fraud Detection Platform)

March 2026

GenAI Genesis 2026 · Team of 4

- Integrated Google Gemini Vision and an LLM service into the fraud-report pipeline for automated invoice and document analysis.
- Built the Next.js risk dashboard (risk overview, flagged transactions, protection score) and the PDF upload flow for the compliance platform.

Multi-Class Painting Classification Model (CSC311 Course Project)

Winter 2026

University of Toronto · Team of 4

- Ran feature-engineering ablations (StandardScaler normalization, TF-IDF vs. numeric-only) across three model families on a 3-class painting-classification task, showing TF-IDF removal dropped Bernoulli Naive Bayes accuracy from 93.5% to 76.7% and shaping the final unscaled, full-feature preprocessing pipeline.
- Analyzed and wrote up feature distributions (numeric, Likert, categorical) across 1,686 survey responses; the team's final neural network reached 94% test accuracy (macro F1 0.94).

mysh (Unix Shell in C)

Winter 2026

CSC209 Course Project · Individual

- Built a Unix shell in C from a course skeleton: builtins (cd, ls, cat, wc, ps, kill), external command execution via fork/execvp, and shell variable expansion.
- Implemented multi-stage pipelines with pipe/dup2, background jobs with SIGCHLD reaping, and SIGINT handling.

Tango (Reinvented Job Matching Platform)

February 2026

DeerHacks V 2026 · Team of 3

- Implemented the core swipe/match feature (role-based discover queue, swipe UI, recruiter inbox) and integrated Auth0 authentication on a Next.js 15 frontend.
- Built a real-time chat feature with a matches sidebar and unread-message badges, and migrated image uploads from filesystem storage to PostgreSQL.

Paint Clone (Java Desktop GUI)

1 month

Team of 4

- Led a team of 4 to build a Windows Paint clone with extended features in Java, applying MVC design patterns and Scrum methodology.
- Managed sprint planning and role assignment across the project lifecycle.

WORK EXPERIENCE

Crew Member

June 2026 – September 2026

Chipotle Mexican Grill

- Ran a high-volume, fast-paced kitchen line, building strong time management and prioritization skills while delivering friendly, consistent customer service during peak rush periods.

Math, Physics & Chemistry Tutor

September 2024 – 2026

Self-employed

- Provided personalized academic support to high school students; adapted explanations to individual learning gaps rather than following fixed curricula.

Customer Service Associate

Summer 2025

Korean Supermarket

- Developed conflict resolution and negotiation skills in a high-volume customer-facing environment.

LEADERSHIP

Finance Associate

2026 – Present

UTM Mathematical & Computational Sciences Society (MCSS)

- Draft and process audit forms for the society's financial records as part of a volunteer executive team representing all full-time Mathematical & Computational Sciences majors at UTM.
- Source and secure sponsors for department events and programming.

VP Finance

2025 – 2026

UTM Salsa & Bachata Club

- Managed the club's annual budget and financial operations for a year, overseeing event expenses, sponsorships, and financial reporting.

ADDITIONAL

Languages: Fluent in English and Korean

Presentations: Weekly presentation experience over three years; comfortable communicating technical concepts to varied audiences.